

The gut contractile organoid: a novel model for studying the gut motility regulated by coordinating signals between interstitial cells of Cajal and smooth muscles

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Abstract

The gut undergoes peristaltic movements regulated by intricate cellular interactions. However, they have poorly been explored due to a lack of model system. We here developed a novel contractile organoid that is derived from the muscle layer of chicken embryonic hindgut. The organoid contained smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs; pacemaker) with few enteric neurons, and underwent periodic contractions. The organoid formed by self-organization with morphological arrangements of ICCs (internal) and SMCs (peripheral), allowing identification of these cells in live. GCaMP- Ca^{2+} imaging analyses revealed that Ca^{2+} transients between ICC- ICC, SMC-SMC or SMC-ICC were markedly coordinated. Pharmacological studies further showed that gap junctions play a role in ICC-to-SMC signaling, and also possible feedback from SMC's contraction to ICC's pace-making activities. In addition, two organoids with different rhythm became synchronized when mediated by SMCs, unveiling a novel contribution of SMCs to ICC's pace-making. The gut contractile organoid developed in this study offers a useful model to understand the mechanisms underlying the rhythm coordination between/among ICCs and SMCs during gut peristaltic movements.

eLife assessment

This **valuable** study reports the development of a novel organoid system for studying the emergence of autorhythmic gut peristaltic contractions through the interaction between interstitial cells of Cajal and smooth muscle cells. While the utility of the organoids for studying hindgut development is well illustrated by showing, for example, a previously unappreciated potential role for smooth muscle cells in regulating the firing rate of interstitial cells of Cajal, some of the functional analyses are **incomplete**. There are some concerns about the specificity and penetrance of perturbations and the reproducibility of the phenotypes. With these concerns properly addressed, this paper will be of interest to those studying the development and physiology of the gut.

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Introduction

In the gut, ingested material is conveyed properly along the gut axis by the gut movements called peristalsis, which is recognized as wave-like propagation of a local constriction. The physiology of gut peristalsis has extensively been studied in adults, where the peristalsis plays pivotal roles in effective transportation and digestion/absorption of inter-luminal contents. And many cases of gut-related pathology are associated with peristaltic dysfunctions. The gut peristaltic movements are achieved by intricate regulations of intercellular functions among multiple cell types. At the site of origin of peristaltic waves (OPW), a local constriction emerges along the circumferential axis, and this is soon followed by a progressive wave of the contraction along the gut axis. During these processes, the circumferential constriction demands multiple SMCs to achieve simultaneous contraction. However, due to the complex structure of the gut, how such synchronization/coordination in SMCs is regulated remains largely undetermined.

It is known in vertebrates that the embryonic gut undergoes peristaltic movements even without experience of food intake. The embryonic gut therefore serves as a powerful model to study the intrinsic mechanisms underlying the peristalsis contrasting with the adult gut where ingested content influences the gut motility increasing complexity in analyses. We have recently reported using chicken embryos that sites of OPW are randomly distributed along the gut axis at early stages, and they later become confined to specific sites, and this confinement of OPWs enables rhythmic and patterned peristaltic movements (Shikaya et al., 2022). One of the long-standing and important questions is how the synchronized/coordinated contraction is achieved.

In a long history of gut motility studies, it has been known that enteric nervous system (ENS), smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs) play important roles in peristaltic movements (Barajas-Lopez et al., 1989; Camborova et al., 2003; Chevalier et al., 2020; Huizinga et al., 1995; Kito et al., 2005; Liu et al., 1998; Rumessen & Thuneberg, 1996; Sanders et al., 1991; Takaki, 2003; Takayama et al., 2002; Thomsen et al., 1998). It has widely been accepted that: 1) at early embryonic stages, gut movement/peristalsis does not require ENS activity, 2) ICCs serve as a pacemaker dictating their rhythm to SMCs, 3) the contraction is executed by SMC and not by ICCs, since thick fibers of myosin are found solely in SMCs. A series of elaborate electrophysiological studies showed that slow waves (a type of changes in membrane potential characteristic of gut movements) occur intrinsically in ICCs but not in SMCs, and that these slow waves lead to voltage-dependent Ca^{2+} influx evoking action potential, which is somehow transmitted to SMCs to execute gut contraction in register with ICC's pace-making rhythm (Baker et al., 2021; Torihashii et al.,

2002). The knowledge that ICCs act as a pacemaker was supported by compelling evidence obtained by c-Kit-deficient mouse mutants (W/W^V), in which ICC differentiation was severely affected leading to a failure of gut peristalsis (Huizinga et al., 1995; Torihashi et al., 1999). However, it remains largely unexplored how the intrinsic/spontaneous rhythm in a single ICC becomes synchronized among multiple ICCs which constitute intricate networks in the gut. It has also been under debate to what extent the gap junction contributes to cell-cell communications between ICCs, SMCs or ICC-SMC. One reason is that it has been difficult to stably maintain SMCs and ICCs in cell culture condition, and also to distinguish ICCs from SMCs in the living gut (these two types of cells originate from the same progenitors of splanchnopleural mesoderm during development). Thus, a novel model system has been awaited to circumvent these obstacles. Recently, it was reported that differentiation states of mouse hindgut-derived cells were successfully maintained for a relatively long period of time in a serum-free culture medium (Wang et al., 2018). In that study, many types of gut-derived cells including not only ENS, ICC, SMCs but also glial cells and serosa were observed, and this large and complex cell mass was seen to undergo rhythmic contractions *in vitro*.

It has widely been appreciated that organoids can serve as a powerful models and tools to circumvent such obstacles of organ complexity. Their relatively simple structures allow analyses at higher resolution than *in vivo*, and also permit analyses of cell behaviors in three-dimensional (3D) environment, which might be different from behaviors *in vitro* behaviors confined to two dimensions. In the current study, we have developed a novel organoid called “gut contractile organoid” by culturing chicken hindgut-derived cells in a serum-free medium. The gut contractile organoid undergoes periodic contractions, and it is essentially composed of ICCs and SMCs, the former residing centrally whereas the latter peripherally, allowing distinction between the two cell types in living organoids. These advantages enabled GCaMP-live imaging of Ca^{2+} dynamics and revealed coordinated oscillations of Ca^{2+} transients between ICC-ICC, SMC-SMC and SMC-ICC. Pharmacological studies further suggested that gap junctions play a role in an ICC-to-SMC signaling, and also a possible feedback from SMC’s contractions to ICC’s pace-making activities. In addition, by regarding a single organoid as an oscillator unit and by placing oscillators separately in a hydrogel mold, we found that two oscillators with different rhythm become synchronized when mediated by SMCs, supporting the notion of SMC’ contribution to ICC’s pace-making. The gut contractile organoid developed in this study must be useful to unveil the intrinsic mechanisms underlying the rhythm coordination between/among ICCs and SMCs during peristaltic movements in the embryonic gut.

Results

Spheroids were formed from muscle layer-derived cells of the embryonic hindgut

To develop a culture condition that would facilitate analyses of gut motility, we dissected the muscle layer (also called *tunica muscularis*) from the hindgut of E15 embryos by removing the serosa and intestinal epithelium (mucosa) (Fig. S1 [↗](#)). The isolated muscle layer was dissociated into single cells to prepare 5.0×10^5 cells per culture dish. We started analyses with a culture condition with FBS-free medium and Matrigel as substrate as previously described for cultures in mice of gastrointestinal cells including serosa (Wang et al., 2018). We tested three kinds of FBS-free media: DMEM/Ham’s F-12, Ham’s F-12, and Neurobasal media (see Materials and Methods). When cultured in DMEM/Ham’s F-12 or Ham’s F-12, dissociated cells formed very small aggregates containing several cells at day 1, the morphology of which did not change significantly until day 5 (Fig. 1A [↗](#)). In clear contrast, when cultured in the Neurobasal medium, cells formed clusters that were interconnected by elongated cells with neighboring clusters as early as day 1. These clusters grew as larger aggregates by day 3, and became spherical by day 5. Such spheroids were not observed in the conditions with Ham’s F-12 or DMEM/Ham’s F-12. We also tested different

substrates, Poly-Lysine or collagen for dish coating with the Neurobasal medium, but neither one yielded spheroid. To know how the spherical aggregates were formed under the condition of Neurobasal medium and Matrigel, we took time-lapse images at two-hour intervals. Originally sparse clusters that were loosely connected by extended cells merged each other forming progressively larger clusters (**Fig. 1B** [↗](#), Movie 1). In the following experiments, we focused on the spheroids formed under the condition of Neurobasal medium and Matrigel coating.

The gut muscle layer-derived spheroids displayed periodic contractions

The spheroids underwent reiterative contractions at day 3, and these contractions were observed at least until day 7 of culture (**Fig. 2A** [↗](#), Movie 2). Time-lapse imaging of the contractions combined with quantitative assessments by ImageJ (see Materials and Methods) showed that the average frequencies of contractions in each cluster/spheroid at day 3, day 5, and day 7, were 3.32, 3.04, and 2.97 contractions/min, respectively (**Fig. 2B** [↗](#)), which largely corresponded to the rhythm of the E16 chicken embryonic intact gut (Chevalier et al., 2019).

The contracting spheroid was composed of ICCs and SMCs

To determine the cell types comprising the spheroids, we performed immunostaining with antibodies for the chicken c-Kit protein (for ICCs; Yagasaki et al., 2022) and α SMA (for SMCs). It is known that ICCs and SMCs are derived from common progenitors which are $c\text{-Kit}^+/\alpha\text{SMA}^+$, and that after differentiation ICCs and SMCs are $c\text{-Kit}^+/\alpha\text{SMA}^-$ and $c\text{-Kit}^-/\alpha\text{SMA}^+$, respectively (Duband et al., 1993; Kluppel et al., 1998). In the clusters at day 3, $c\text{-Kit}^+/\alpha\text{SMA}^-$ signals were detected in the internal region, and $c\text{-Kit}^+/\alpha\text{SMA}^+$ signals at the periphery (**Fig. 3A** [↗](#), Day 3), suggesting that internal cells were differentiated ICCs, whereas peripheral cells were ICC/SMC progenitors. At day 5 onward, $c\text{-Kit}^+/\alpha\text{SMA}^-$ cells and $c\text{-Kit}^-/\alpha\text{SMA}^+$ cells were segregated which were located internally and peripherally, respectively (**Fig. 3A** [↗](#), Day 5), and this spatial segregation of the cells remained unchanged until day 7 (**Fig. 3A** [↗](#), Day 7), the latest stage examined in this study. Infection of RCAS-GapEGFP into a forming spheroid further visualized multipolar ICCs in the internal region (**Fig. 3B** [↗](#)), known to be characteristic of ICC-MY which would normally reside in the layer of myenteric plexus (Mei et al., 2009; Sanders et al., 2014). ICCs of a different type were also recognized, which were elongated, thin, and lining underneath the peripheral SMCs (**Fig. 3A** [↗](#), Day 7). It is possible that they were ICC-IM, known to be embedded in and tightly associated with smooth muscles in the gut (Huizinga et al., 2011; Iino & Horiguchi, 2006). Unexpectedly, the spheroid contained few neural cells (ENS), if any, revealed by anti-Tuj1 antibody (**Fig. S2** [↗](#)).

Since ICCs and SMCs were spatially segregated in the spheroid, we stained with anti N-cadherin antibody (E-cadherin is positive solely in the mucosa/endoderm) (Graham et al., 2017; Grosse et al., 2011). N-cadherin signals were seen in the internal ICCs, and not in the peripheral SMCs (**Fig. 3C** [↗](#)). Thus, it is likely that N-cadherin plays a role in the segregation.

In summary, the hindgut-derived spheroid displays three prominent characteristics: (1) a dominant occupation by ICCs and SMCs with neglectable contribution by ENS, (2) self-organization ability of internal ICCs encapsulated by a thin layer of SMCs, (3) contractions with periodic rhythm. Based on these characteristics, we designated this spheroid as “gut contractile organoid”.

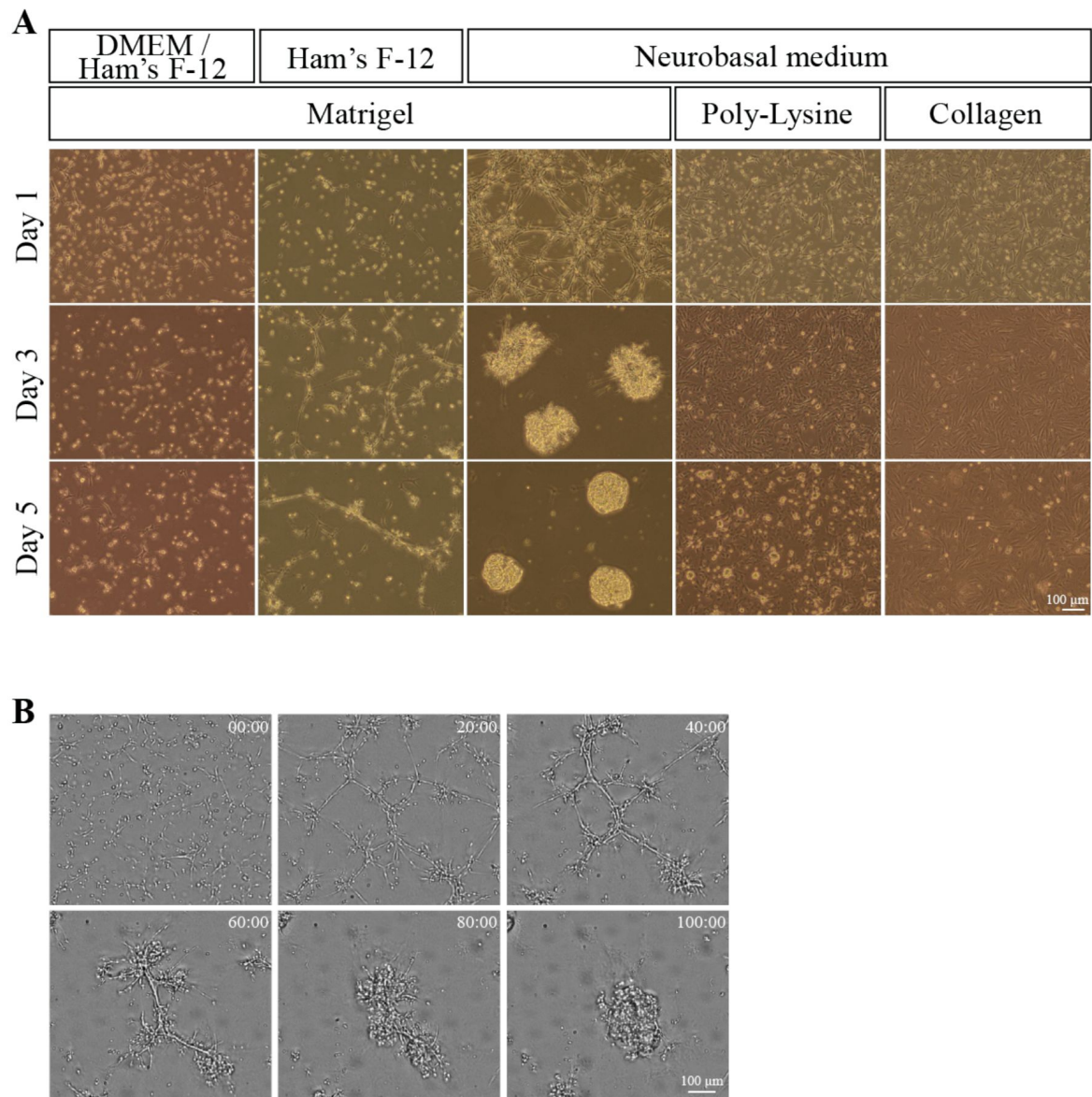


Figure 1

Culture of muscle layer-derived cells prepared from embryonic hindgut.

(A) Culture of muscle layer-derived cells prepared from embryonic hindgut with FBS free-media and substrates. (B) Long-term time-lapse imaging after seeding on Matrigel with Neurobasal media. The images show the ability of these cells to self-assemble at 0, 20, 40, 60, 80 and 100 hours taken from Movie 1. Scale bars: 100 μ m in A, B.

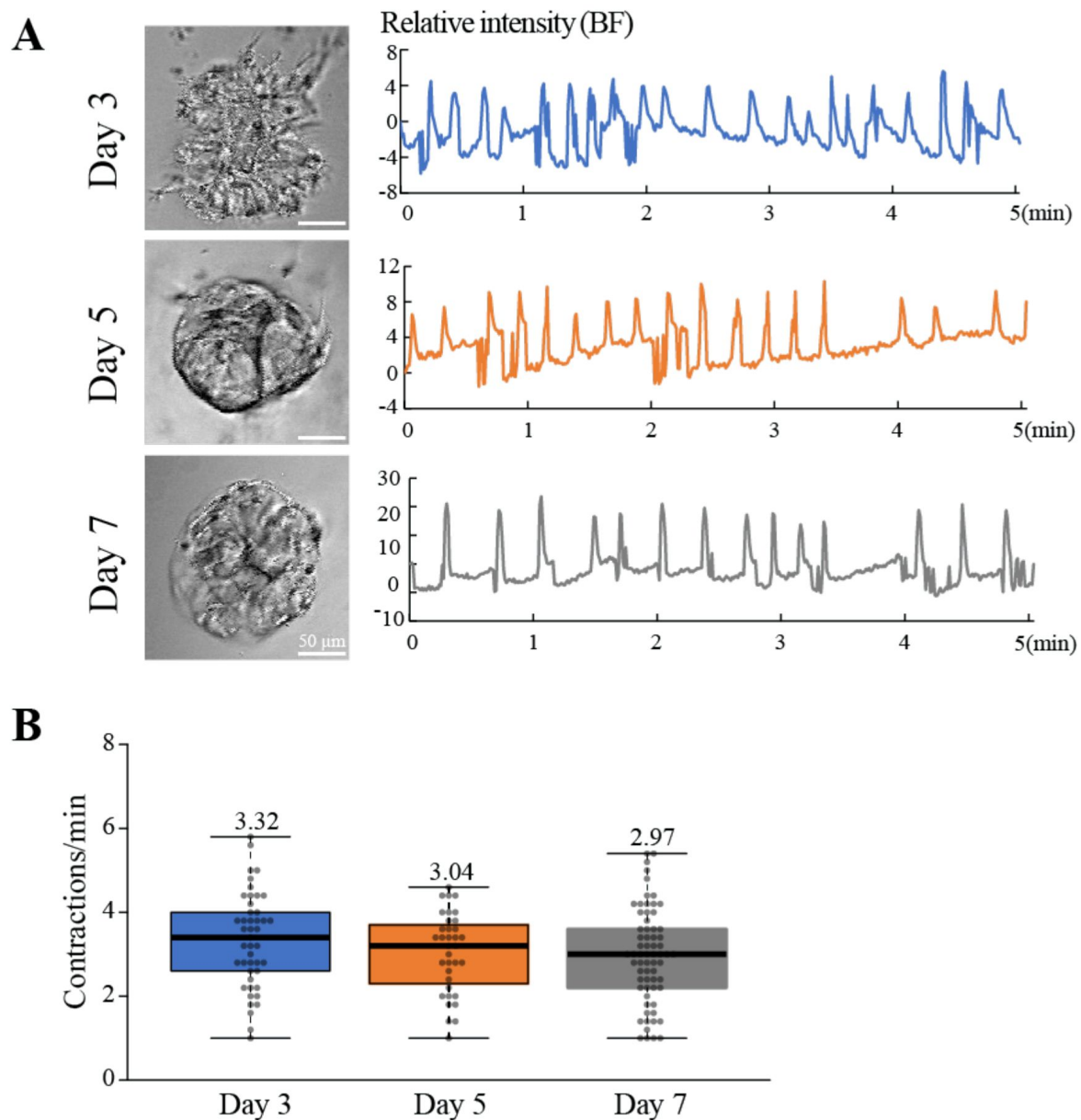


Figure 2

Forming spheroids exhibited reiterated contractions.

(A) Clusters/spheroids on day 3, 5 and 7 exhibited periodic contractions. Graphs show contractions quantified by differences in the intensity of brightfield in sequential images. (B) Frequency of contractions of the cluster/spheroid on day 3-7 (N=day 3: 46, day 5: 35, day 7: 60). The number on each graph shows the average of contraction frequencies. Scale bars: 50 µm in A.

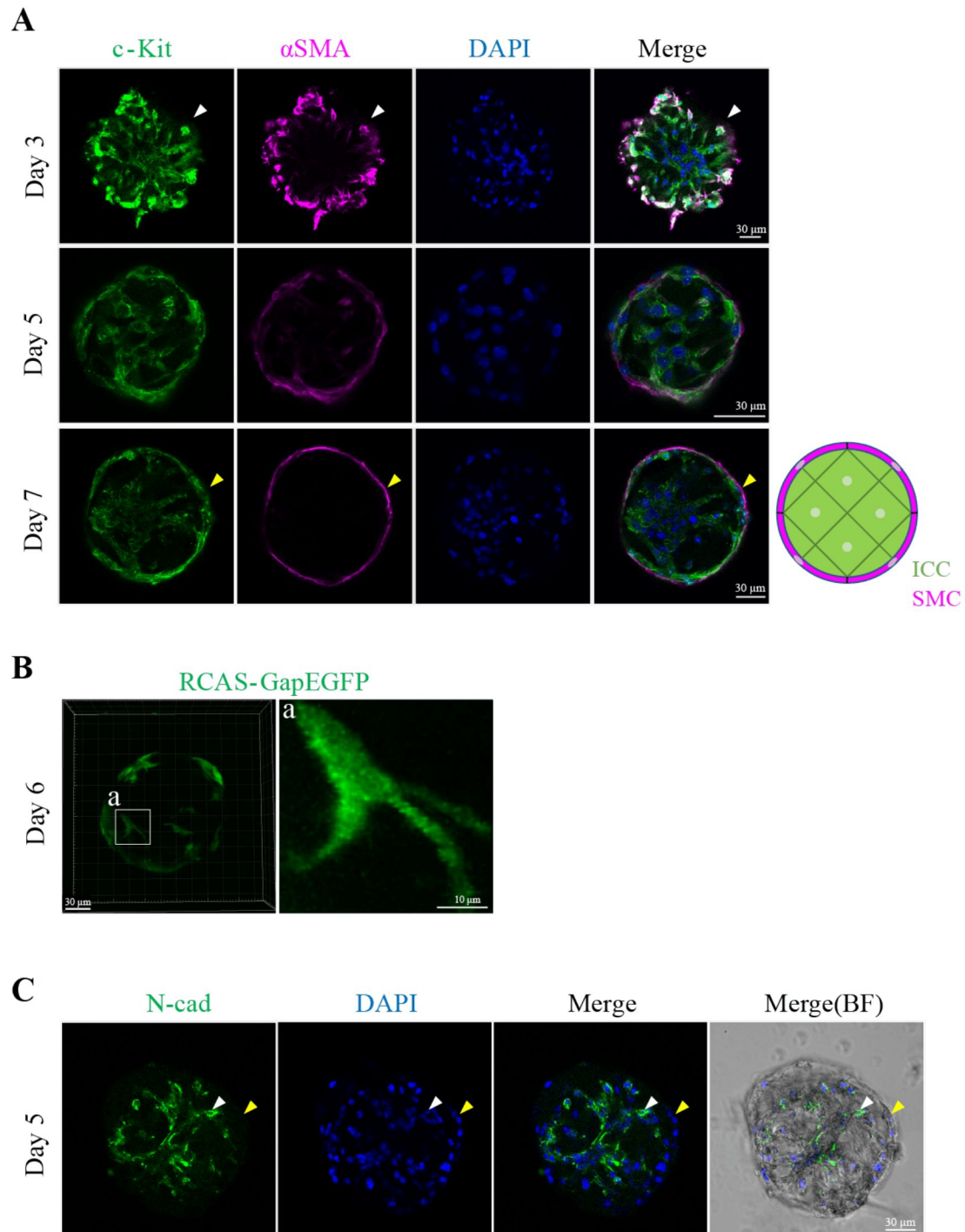


Figure 3

The cluster/spheroid is composed of ICCs internally and SMCs peripherally.

(A) Co-staining with anti-c-Kit- and α SMA antibodies. White arrowheads show co-expression of c-Kit and α SMA at Day 3. Yellow arrowheads indicate a cell expressing α SMA but not c-Kit at day 7. A diagram shows an image of cell arrangement (green: ICC, magenta: SMC) in the spheroid at day 7. (B) Cell morphology in the spheroid at day 6 revealed by RCAS-gapEGFP expression. (C) Staining of spheroids at day 5 with anti-N-cadherin antibody. A white arrowhead shows expression of N-cadherin. A yellow arrowhead shows a cell which does not express N-cadherin in the outer region of the spheroid. Scale bars: 30 μ m in A-C, 10 μ m in a.

Contraction-associated intracellular Ca^{2+} transients were coordinated between ICCs and SMCs in the gut contractile organoid

It has been reported that Ca^{2+} dynamics are important for pacemaker activity in ICCs and contractions of SMCs. Ca^{2+} flows into ICCs via voltage-dependent Ca^{2+} channels and propagates to SMCs, causing the gut muscle contractions (Baker et al., 2021). We therefore investigated the Ca^{2+} dynamics in our gut contractile organoids.

Organoid-forming cells were infected with a RCAS-vector encoding GCaMP6s, a Ca^{2+} indicator that emits EGFP fluorescence in response to Ca^{2+} influx, and mRuby3 as a reporter (Fig. 4A). GCaMP6s-organoids were subjected to time-lapse imaging analyses by confocal microscopy. As expected, the oscillations rhythm of Ca^{2+} transients as a whole organoid was concomitant with that of contractions (Fig. 4B, Movie 4).

Taking advantage of the spatial segregation between ICCs (internal) and SMCs (peripheral) (Fig. 3A), we compared the Ca^{2+} oscillation rhythm among ICCs and SMCs (homotypically) and between ICC-SMC (heterotypically). We set up 3 regions of interests (ROIs) in the central region, with one ROI corresponding to one cell, and captured the Ca^{2+} transients. The three ROIs exhibited a synchronous pattern of Ca^{2+} oscillations (Fig. 4C, ICCs). Similarly, Ca^{2+} oscillations in three ROIs in the SMC layer were synchronous (Fig. 4C, SMCs). These data highlight active communications taking place intercellularly within ICCs and SMCs, respectively. We further compared Ca^{2+} oscillations between ICC-SMC. Setting up with one ROI in an ICC, and one in a SMC. Again, the Ca^{2+} rhythm was synchronized between these heterotypic cell types (Fig. 4C, ICC/SMC). With a closer look, a rise of the Ca^{2+} transient in the ICC preceded that in SMC with a time lag (also called latency) of 690 msec in average (Fig. 4C, ICC/SMC), suggestive of a propagation of Ca^{2+} signals from ICC to SMC consistent with previous reports (Baker et al., 2021).

Gap junction plays a role in ICC-to-SMC signaling

Although a series of gut motility studies have proposed an importance of gap junctions, rigid evidence has been limited due to a lack of experimental model. Our gut contractile organoid should prove useful for clarifying the roles of gap junctions in the synchronous motility, since intercellular synchronization was observed between/among identifiable cells (ICCs and SMCs) as shown above. We performed pharmacological assessments using gap junction inhibitors including carbenoxolone (CBX) and 18 β -glycyrrhetinic acid (18 β -GA; Chevalier, 2018; Takeda et al., 2005), two of the most widely used pharmacological inhibitors used to study gap junctions.

Following inhibitor administration into day 7 culture medium, organoids were allowed to rest for 30 min to exclude possible effects by the administration, for example, turbulence of the medium. With CBX or 18 β -GA in the culture medium, the periodic contraction of an organoid was retained with almost the same frequency as the control (Fig. S3, Movie 3). We then examined whether the intercellular synchronization/coordination was affected by the inhibitors. Similarly to Fig. 4C, three ROIs were set either in ICCs or SMCs territories for assessment of Ca^{2+} transients. The same cells in a single organoid were tracked for their Ca^{2+} transients before and after the inhibitor administration. Contrary to our expectation, none of 10 μM , 20 μM and 100 μM concentrations of these inhibitors showed detectable inhibition of the synchronous patterns of Ca^{2+} transients (Fig. 5A, B show 100 μM of CBX, Movie 5).

In contrast, when an ICC and SMC were compared, the preceding rise of Ca^{2+} transient in ICC was abolished, although overall synchronization was retained between these the cells (Fig. 5C). Collectively, while the contribution by gap junction to the periodic contraction and intercellular

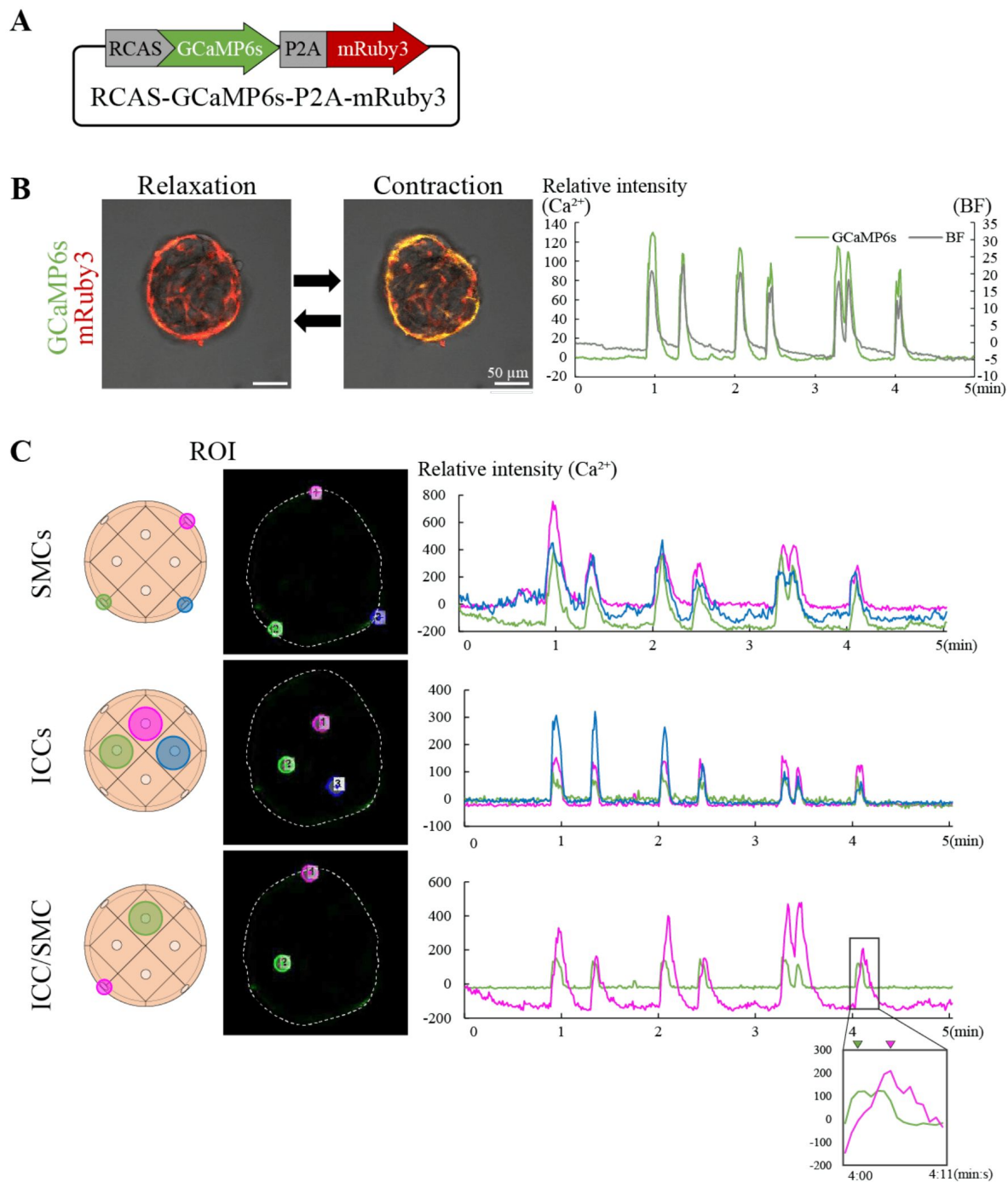


Figure 4

Ca^{2+} imaging of the gut contractile organoid revealed intercellular synchronization.

(A) RCAS-GCaMP6s-P2A-mRuby3 plasmid. (B) Ca^{2+} imaging of gut contractile organoid during relaxation and contraction. Ca^{2+} dynamics (green) and contraction (grey) of gut contractile organoid. (C) Simultaneous measurement of intercellular Ca^{2+} dynamics between ICC-ICC, SMC-SMC or ICC-SMC. Three or two ROIs with Ca^{2+} signal-positive cells were set in a single organoid. Graphs show Ca^{2+} dynamics in the ROIs. Magnified view shows that a rise of Ca^{2+} signal in ICC (green) preceded that in SMC (magenta). Scale bar: 50 μm in B.

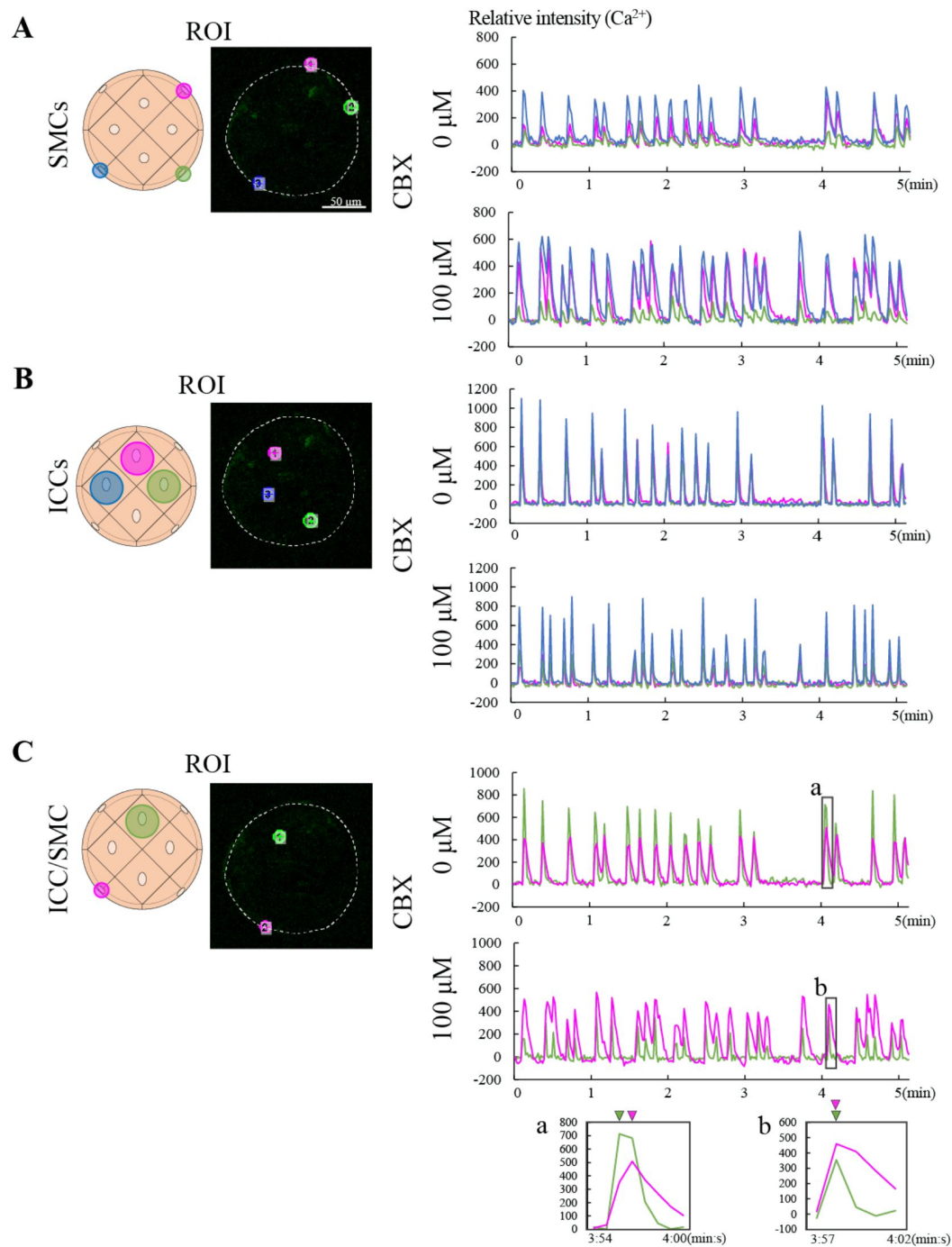


Figure 5

Gap junction inhibitor exerted limited effects on the synchronization of Ca^{2+} dynamics.

GCaMP6s-expressing organoids were cultured with CBX, and three or two ROIs were assessed for their Ca^{2+} synchronization. A single organoid was subjected to the assessment before (0 μM ; control) and after administrations of 100 μM CBX. The synchronization was unaffected between (A) SMC-SMC and (B) ICC-ICC, but that in (C) ICC-SMC was partially affected. Magnified view shows that the preceding rise of Ca^{2+} in ICC (green) before CBX was abrogated after CBX. Scale bar: 50 μm in A.

synchronization in the organoid is relatively limited, the ICC-to-SMC signals require gap junction-mediated communications, at least partly.

Blebbistatin abolished organoidal contractions and oscillatory patterns of Ca^{2+} transients

Toward searching for factors that regulate the coordination between/among ICCs and SMCs, we tested blebbistatin, a specific inhibitor of myosin II, which was expected to cease organoidal contractions. Experimental procedures were similar to those for gap junction inhibitors. We found that blebbistatin ceased periodic contractions of organoids in a concentration-dependent manner, with 10 μM ceasing it completely (Fig. 6A, Movie 6). We examined Ca^{2+} transients in these contraction-inhibited organoids. Markedly, periodic Ca^{2+} transients not only in SMCs, but also in ICCs were extinguished, resulting in no/little synchronous Ca^{2+} patterns among and between ICCs and SMCs (Fig. 6B-E). Although the possibility is not excluded that blebbistatin directly inhibited non-muscle myosin II in ICCs, these findings imply that the contractility feeds back to ICCs to generate their periodic rhythm.

Inter-organoidal coordination was mediated by SMCs

During analyses with our novel organoids, we found that they easily fuse to each other, suggesting that organoids grow by progressive fusion (Fig. 7A, Movie 7). This also raised the possibility that the synchronous Ca^{2+} transients among organoid-constituting cells (Fig. 4C) might be a consequence of phasic coordination upon the fusion of multiple organoids that had shown different oscillatory phases. To test this possibility, we transferred two organoids in a petri dish, and allowed them to fuse. Before the fusion, Ca^{2+} oscillatory rhythm was indeed out of phase in the two organoids (Fig. 7B, Movie 8). Markedly, upon fusion by 24 hours, their rhythm became in phase/synchronous (Fig. 7B, Movie 8).

Since we noticed that cellular protrusions were often observed around the time of organoidal fusion, we reasoned that these cellular processes would mediate the fusion and its subsequent synchronization of Ca^{2+} transients (Fig. 7C). To test this, we developed a 3-well hydrogel with narrow channels connecting the wells (Fig. 8A). Organoids were placed separately in each well, which prevented the fusion of organoidal bodies, but allowed extension of cellular processes through the narrow channels and contact with each other (Fig. 8B). After 72 hours of placement of organoids, cellular processes as well as several cell bodies were present within the channel, and Ca^{2+} transients became synchronized among the three organoids (Fig. 8C, Movie 9). However, with careful examination, we also noticed that organoid-derived cells crawled out from the wells to cover the top surface of the hydrogel, connecting the three organoids (Fig. 8D). Thus, another possibility was raised that these crawled out cells would mediate the inter-organoidal synchronization.

To test this possibility, we prepared a 3-well hydrogel in which the three wells were disconnected (no channels) so that organoid-derived cellular processes were not able to connect each other (Fig. 8E). By 3 days of culture, the top surface of the hydrogel was indeed covered by cells in a similar way to Fig. 8D. These cells were positive for αSMA but negative for c-Kit (Fig. 8F), indicating that they were SMCs that were somehow detached and crawled out from the peripheral layer of the organoids. Importantly, coincided with the top-coverage by the SMCs, the three organoids in the disconnected wells displayed synchronized Ca^{2+} transients, highlighting the role of SMCs in mediating coordination between organoids (Fig. 8G, Movie 10). Gap junction inhibitor yielded no/little effects on the Ca^{2+} transient coordination (Fig. 8H, Movie 11).

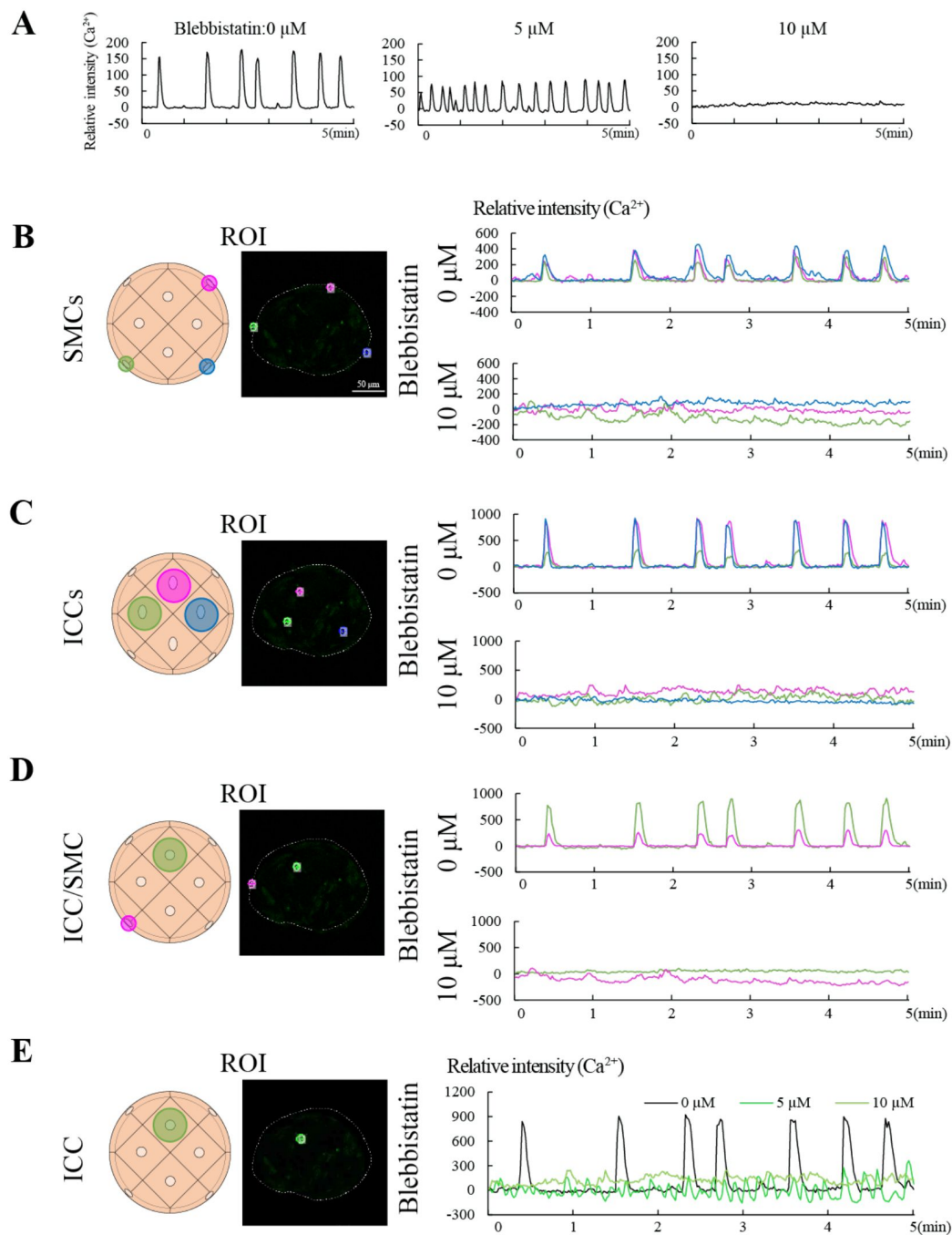


Figure 6

The organoidal contraction is important for Ca^{2+} dynamics in ICCs.

GCaMP6s-expressing organoids were cultured with Blebbistatin. (A) Ca^{2+} oscillation was extinguished in the entire organoid by administration of Blebbistatin. (B-D) Ca^{2+} dynamics in (B) SMC-SMC, (C) ICC-ICC and (D) ICC-SMC. Three or two ROIs were assessed for their Ca^{2+} synchronization. A single organoid was subjected to the assessment before (0 μ M; control) and after administrations of 10 μ M Blebbistatin. (E) Ca^{2+} transients in a single ICC in 0 μ M, 5 μ M, and 10 μ M. Scale bar: 50 μ m in B.

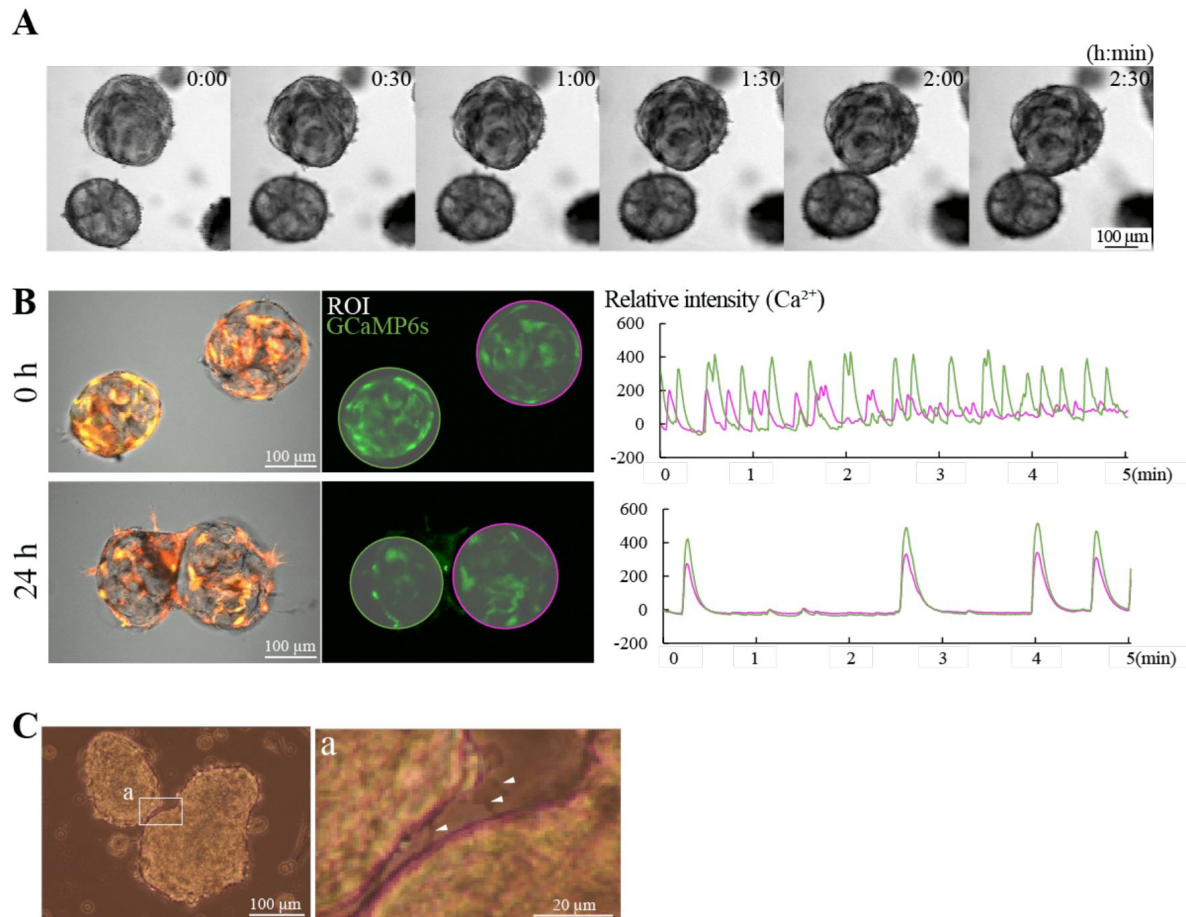


Figure 7

Ca^{2+} dynamics in multiple gut contractile organoids undergo synchronization upon their fusion.

(A) Time-lapse imaging of organoidal fusion. (B) When two organoids that originally showed independent rhythm of Ca^{2+} fused to each other, their rhythm became synchronized after fusion (24 h). (C) Protrusions between two neighboring organoids. White arrowheads show three protrusions from the left organoid. Scale bars: 100 μm in A-C, 20 μm in a.

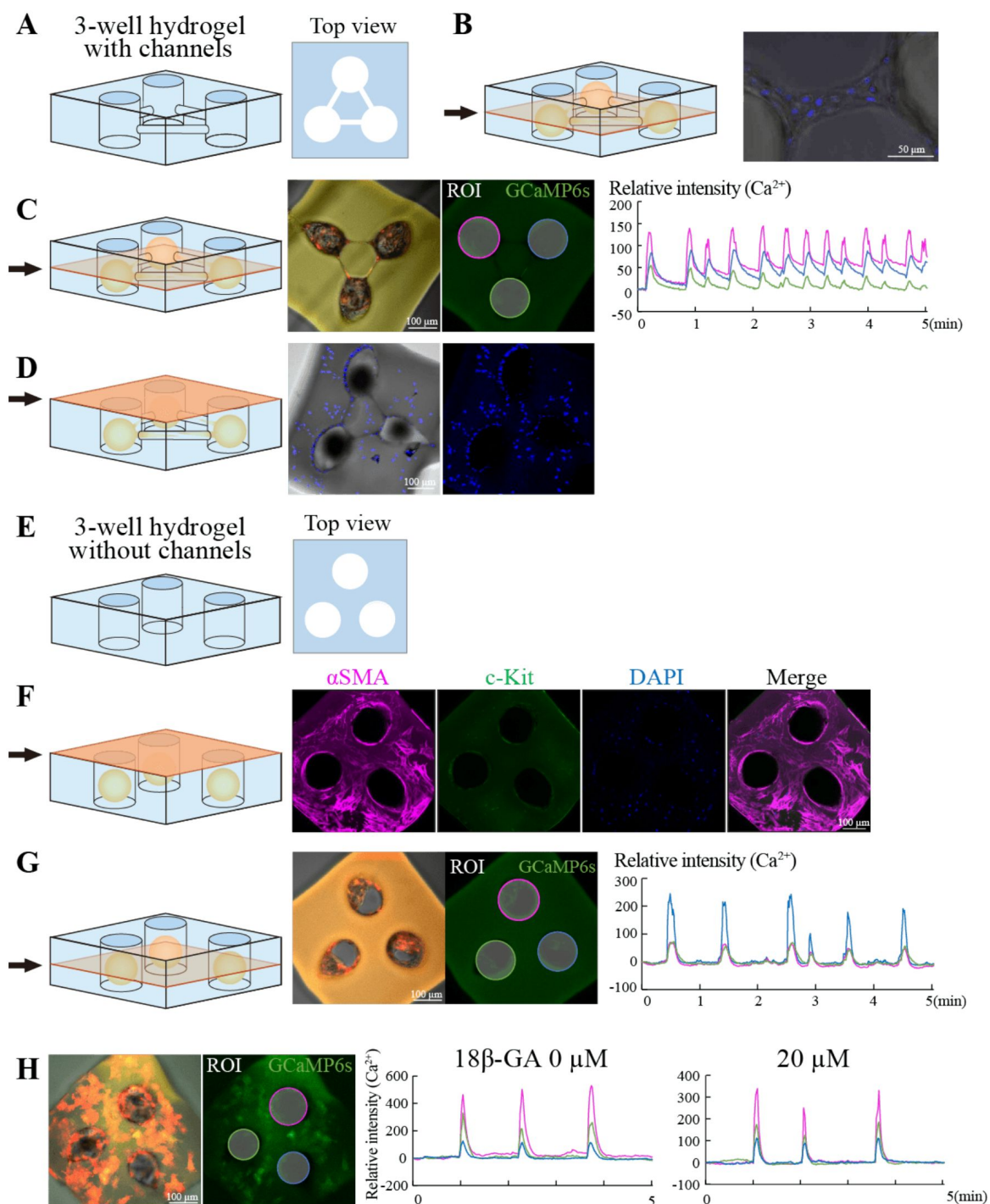


Figure 8

Smooth muscle cells mediate the Ca²⁺ synchronization between organoids.

(A) Diagram of 3-well hydrogel in which an organoid placed in each well. This gel mold does not allow organoidal bodies to fuse to each other, but allow them to extend/migrate protrusions/cells through a narrow channel connecting the wells (B). (C) After 3 days, three organoids displayed synchronization of Ca²⁺ dynamics. (D) Some organoid-derived cells crawled out from the wells and covered the top surface of the hydrogel, resulting in bridging the three unfused organoids. (E) Diagram of 3-well hydrogel *without* channels (F) Surface-covering cells were SMCs (αSMA positive and c-Kit-negative). (G) In the hydrogel without channels but with surface-covered SMCs, Ca²⁺ dynamics in the 3 organoids were synchronized. (H) The Ca²⁺ synchronization shown in (G) was not altered by 18β-GA. Scale bars: 50 μm in B, 100 μm in C-H.

Discussion

We have developed a novel gut contractile organoid, which displays several unique characteristics: 1) it undergoes periodic contractions, 2) differentiation states of ICCs (c-Kit⁺/aSMA⁻) and SMCs (c-Kit⁻/aSMA⁺) are maintained at least until day 7 in the organoid, 3) the organoid is composed essentially of two types of cells, ICCs and SMCs, with few ENS cells, if any, 4) ICCs (internal) and SMCs (peripheral) can be distinguished for their localization in a living organoid, allowing 5) GCaMP-visualization of Ca²⁺ transients and assessments of cell interactions between and among ICCs and SMCs. These characteristics circumvent, at least partly, obstacles that have hampered analyses in the research of gut peristalsis, such as unstable differentiation state of ICCs and SMCs in cultures, and difficulties in identifying these cells in living preparations. In studies of gut movements, how ICCs generate/maintain their periodic rhythm and how ICCs and SMCs interact with each other have been long-standing questions, and our organoids offer powerful advantages to address these and to understand the cellular mechanisms underlying the gut contractions/peristaltic motility.

Contrasting with many cases in organoid studies that have aimed at a *maximum* recapitulation of the intact organ, our gut contractile organoid is composed of a (probably) *minimum* number of cell types that suffice the generation and/or maintenance of rhythmic contractions, allowing high-resolution analyses at the cellular level. For example, adding ENS components to our organoid would allow the clarification of the role of ENS in the gut contraction. While the majority of internal cells in the organoid are ICCs (c-Kit⁺/aSMA⁻), the possibility cannot be excluded that platelet-derived growth factor receptor- α positive (PDGFR α ⁺/c-Kit⁺/aSMA⁻) cells, another type of interstitial cells (fibroblast-like cell) known to mediate neural activity to SMCs in the mouse gut (Sanders et al., 2024; Sanders et al., 2016), are included in our organoid. Available antibody against the chicken PDGFR α protein is awaited.

Coordinated Ca²⁺ transients in ICC/SMC populations in the gut contractile organoid

Measurement and quantification analyses with ICCs and SMCs that are identifiable in the living organoid revealed exquisite coordination of Ca²⁺ transients/oscillation homotypically between ICC-ICC and SMC-SMC and heterotypically between SMC-ICC (**Fig. 4C**). Notably, a rise (upstroke) of Ca²⁺ transient in ICCs precedes that of SMCs with a time lag (also called latency) of 690 msec, suggesting a directed signaling from ICC to SMC. These observations are consistent with previous studies using ICC- and SMC-specific transgenic mice that expressed GCaMP and RCaMP, respectively (Baker et al., 2021). In that study, the authors assessed Ca²⁺ transients in submucosal ICCs (ICC-SM) and compared them with those in their adjacent circular muscles, and showed that the rise of GCaMP signal preceded that of RCaMP with a latency of 56 $\pm$ 14 msec. With these observations, they concluded that ICC-SM signals to its adjacent SMCs. In our study, time-lapse imaging was mostly performed with 700 msec intervals. Further studies with shorter intervals are awaited to know the latency time would be shorter than 690 msec in average.

Contribution of gap junction to ICC-to-SMC signaling

Effects by the block of gap junction by CBX or 18 β -GA were relatively limited in our organoid assays: synchronous patterns of Ca²⁺ transients remained unchanged between SMC-SMC and ICC-ICC. In contrast, signaling from ICCs to SMCs was specifically affected by these inhibitors, in which the preceding rise of Ca²⁺ transient in ICCs was abolished. The contribution of gap junction to the ICC-to-SMC signaling was previously reported in mouse gut, which was from intramuscular ICCs (ICC-IM) to their adjacent circular smooth muscles. However, interpretation of the role of gap junction in ICC-SMC interactions, in general, has been under big debate. Some studies reported

that CBX or 18 β -GA failed to inhibit these interactions or peristaltic motilities (Komuro et al., 1996; Rohr et al., 1998; Schultz et al., 2003), or that electron microscopy did not detect structures of gap junction in longitudinal muscles (Cousins et al., 2003; Daniel & Wang, 1999; Gabella & Blundell, 1981). Contribution of gap junctions to the gut motility appears to be highly variable in different regions of gastrointestinal tract, for example, stomach versus colon (Iino et al., 2007; Yang et al., 2012). And a detection of gap junction structures or mRNA/protein of Connexins (Cx; a component of gap junction) does not necessarily mean that the gap junction is functional. Dominant negative form of gap junction might be useful. Currently, ICC- or SMC-specific gene manipulations in our organoid are not available, and further studies are awaited. It is intriguing that the coordination between SMC-SMC and ICC-ICC was unaffected by the inhibitors in our organoids. The possibility cannot be excluded that gap junctions that are not inhibited by CBX or 18 β -GA might operate. However, since these two inhibitors have popularly been used to block a wide range of gap junctions, alternative possibility arises that cell communications particularly between SMC-SMC are mediated by other mechanisms than gap junctions, including mechanical signals as recently shown for coordinated Ca²⁺ signaling in cardiac muscle cells (Fukui et al., 2021).

Possible feedback from SMC's contractility to ICC's oscillatory rhythm

Blebbistatin prevented the contraction of the organoids, which was concomitant with abrogation of Ca²⁺ transients in ICCs. Since it has been reported that thick myosin fibers necessary for the contraction are found only in SMCs but not in ICCs (Gherghiceanu & Popescu, 2005; Rumessen & Thuneberg, 1991, 1996; Sun et al., 2006), it is likely that ICCs do not have contractile ability. Our observations therefore promote the possibilities that ICC's pace-making activity requires its own non-muscle myosin II, and/or mechanical feedback by contracting SMCs. Currently, techniques that distinguish between these possibilities are unavailable. Nevertheless, if it is the latter case, it would indicate a signaling from SMC to ICC, the direction of which is unprecedented, since previous literatures had shown that it was solely an ICC-to-SMC signaling process that dictated the rhythm of SMC's contraction. We presume that the SMC's contribution to ICC's pace-making is highly possible since this notion is further supported by other findings obtained in this study showing that SMCs mediate inter-organoidal phase coordination (**Fig. 8** [↗](#)) as more discussed below.

It is of note that the reciprocal interactions between pace-making cells and their effectors have been reported in studies of neural circuit establishment. Spontaneous activities emerging in motor neurons during embryogenesis are regulated by feedback from their governing muscle's contractions so that motor neurons get organized to display coordinated and stable oscillatory activities (Zeng et al., 2021). Thus, it is tempting to speculate that during early gut peristalsis, ICCs might receive feedback from their governing SMCs to generate more stable coordination of pace-making activity among ICCs. Indeed, we have previously reported that in the very early embryonic gut, origins of peristaltic wave (OPW) are randomly distributed along the gut axis, but they later become confined to specific sites showing that ICCs undergo more stable and coordinated pace-making (Shikaya et al., 2022).

Interactions between ICCs and SMCs in the gut contractile organoid

The organoid developed in this study was derived from the muscle layer (also called *tunica muscularis*) of chicken E15 hindgut devoid of mucosa and serosa, in which myenteric ICCs (ICC-MY), intramuscular ICCs (ICC-IM), and submucosal ICCs (ICC-SM) are localized in a way similar to those in mice, shown by staining with antibody against the chicken c-Kit protein (Yagasaki et al., 2022). In the current study, c-Kit antibody-staining showed two types of cells in morphology in the gut contractile organoid: one is multipolar and N-cad-positive ICCs located centrally, and the other is ICCs that are thin in shape, N-cad negative, and lining beneath the most external layer of SMCs.

Based on the knowledge obtained in studies with mammalian species that ICC- MY are multipolar whereas ICC-IM are bipolar and tightly associated with adjacent SMCs (Huizinga et al., 2011; Iino & Horiguchi, 2006), it is conceivable in our organoids that the central cells are ICC-MY, and the peripherally lining ones are ICC-IM. In addition, it has been reported in mice that signaling from ICC-IM to SMCs is gap junction-dependent. Collectively, our observations that gap junction-dependent signal found between ICCs (central ICC) and SMCs in the organoid (**Fig. 5C**) could be interpreted as follows: ICC-MY (central) signals to ICC-IM (second-most peripheral), which in turn signals to the external SMCs mediated partly by gap junction. Such sequences of signaling (ICC-MY to ICC-IM to SMCs) has also been proposed in the intact gut in mammals, although rigid evidence has not been known. Direct comparison of Ca^{2+} transients between the ICC-IM-like cells and SMCs in our organoid was technically very difficult since these two cells were both thin and tightly associated each other.

The gut contractile organoid provides a useful model and tool for studying phasic coordination of oscillatory rhythm

To understand the gut peristaltic movements, which reiterate at specific sites along the gut axis (Shikaya et al., 2022), deciphering the mechanisms underlying the coordination/synchronization of oscillators among multiple cells is critical. Exploiting the finding obtained in the current study that multiple organoids easily fuse each other in vitro, we have found that two organoids with different oscillatory rhythm eventually coordinate their phases upon the fusion (**Fig. 7**). This suggests the ability of ICCs to adjust their rhythm to their neighbors. During identification of the rhythm-adjustment mediators using the 3-well hydrogel, contrasting with our original expectation, SMCs that were crawled out from the organoids and covered the top surface of the hydrogel were able to mediate the rhythm coordination among organoids (**Fig. 8**). Whether cellular thin protrusions connected with each other in narrow channels mediate the coordination remains undetermined. Nevertheless, our findings have revealed a novel role of SMCs in mediating rhythm coordination, and as discussed above, this supports the notion of a SMC-to-ICC signaling. It is unlikely that gap junction plays a major role in such signaling since gap junction inhibitor yielded no detectable effects in the assays using the 3-well hydrogel.

In summary, the gut contractile organoid developed in this study serves as a powerful model to study the establishment and maintenance of oscillatory rhythm (pace- making) and their coordination in the multicellular systems.

Materials and Methods Chicken embryos

Fertilized chicken eggs were obtained from the Yamagishi poultry farms (Wakayama, Japan) and Takeuchi Farm (Nara, Japan). Embryos were staged according to embryonic days. All animal experiments were conducted with the ethical approval of Kyoto University (#202110).

Culture preparation of hindgut-derived cells

A hindgut was dissected from E15 chicken embryos (**Fig. S1A**), and cut into small pieces. After treating with 25 U dispase (Fujifilm Wako, 383-02281) /phosphate buffer saline (PBS: 0.14 M NaCl, 2.7 mM KCl, 10 mM $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$, 1.8 mM KH_2PO_4) at 38.5 °C for 40 min, serosa and intestinal epithelium were removed using forceps. The muscle layer was minced into smaller pieces and treated with 0.2 mg/ml collagenase (Fujifilm Wako, 038-22361) and 0.25 % trypsin/PBS at 37.0 °C for 30 min. The reaction was stopped with 1 % FBS/PBS followed by centrifugation at 800 rpm for 3 min. The pellet was resuspended and washed in PBS followed by centrifugation. The pellet was resuspended in culture medium, and 5.0×10^5 cells were plated in a glass bottom dish (Matsunami, D11130H) treated with Matrigel (Corning, 354248) prepared in advance at 38.5 °C for 20 min before use. Poly-lysine and collagen-coated dishes were purchased (Matsunami, D11131H, D11134H). D-MEM /Ham's F-12 (Wako, 048-29785), Ham's F-12 (Wako, 087-08335) and Neurobasal

medium (Gibco, 21103-049) with 1× B-27 supplement (Gibco, 17504044) and 0.5 mM L-glutamine were tested. After seeding, time-lapse images were obtained with CM20 (Evident) under 5% CO₂ and 38.5°C with 2-hour intervals.

Assessment of contraction in the spheroid/cluster

The time-lapse images were obtained by confocal microscopy (Nikon, A1R) under 5% CO₂ and 38.5°C. The bright field images were converted to the sequence of Tiff images. Using ImageJ (National Institutes of Health, USA), these sequences were divided into foreground and background automatically (Auto Threshold, mode “default”) and graphed by Plot Z-axis profile.

Immunocytochemistry

An organoid was fixed in acetic acid/ethanol (1:5) for 10 min at room temperature (RT), and washed in PBS for 10 min at RT. The specimens were permeabilized in 0.1% Tween-20 in PBS for 10 min at RT, followed by washing in PBS for 10 min at RT. After blocking with 1% blocking reagent for 1 hour at RT, specimens were incubated overnight at 4°C with dilution of 1:300 anti-c-Kit (Yagasaki et al., 2022), 1:300 Tuj-1 (RSD, MAB1195), 1:400 anti-αSMA antibody (Sigma-Aldrich, A5228), and/or 1:200 anti-N-cadherin antibodies (TAKARA, M110) in 1% blocking reagent (Roche, 1096176)/PBS. Following three times washing in PBS for 10 min each at RT, specimens were incubated for 1.5 hr at RT with 1:500 anti-rabbit IgG(H+L)-Alexa 488- conjugated antibody (Donkey; Invitrogen, A21206), anti-mouse IgG2a-Alexa 568- conjugated antibody (Goat; Invitrogen, A21134), anti-rat IgG (H+L)-Alexa 488- conjugated antibody (Goat; Invitrogen, A11006) and 1:2000 DAPI. After washing three times in PBS for 10 min at RT, fluorescent images were obtained using the Nikon A1Rconfocal microscope.

Plasmids

pAAV-hSynapsin1-GCaMP6s-P2A-mRuby3 was purchased from Addgene (112007). Full-length cDNA of GCaMP6s was PCR-amplified:

forward 5'- GCGTACCACTGTGGCATCGATGCCACCATGGGTTCTCA -3', reverse 5'- GCCCGTACATCGCATCGATTACTTGTACAGCTCGT -3'.

The retroviral vector RCAS-EGFP was digested with ClaI to remove EGFP, into which a DNA fragment was inserted by In-Fusion HD Cloning Kit (TAKARA) to produce RCAS-GCaMP6s-P2A-mRuby3. RCAS-GapEGFP is as previously described (Murai et al., 2015)

Preparation of retroviral vector particles

RCAS-GCaMP6s-P2A-mRuby3 was transfected into the chicken fibroblast line cells DF1 using Lipofectamin 2000 (Invitrogen). Transfected cells were cultured in 10 cm culture dish until confluent. The supernatant of transfected DF1 was collected for viral precipitation, from which retroviral particles were prepared using Retro-ConcentinTM Virus Precipitation Solution (SBI, RV100A-1).

Intracellular Ca²⁺ imaging in the gut contractile organoid

At day 2 or day 3 of cell culture, 10 mg/ml polybrene solution (final concentration 4 μg/ml, Nacalai, 12996-81) and 20 μl opti-mem with above mentioned virus particles were added to the culture medium to transfect GCaMP6s into organoid-forming cells. The medium was replaced with fresh medium at day 5 and Ca²⁺ imaging was performed at day 7. Time-lapse images were obtained using the confocal microscope (Nikon, A1R) with 700 or 450 msec intervals, and exported intensity of each region of interest (ROI) into an excel file by time measurement (Nikon, NIS-Elements). From the excel file, the graphs were plotted with the fluorescence intensity by setting the beginning of the measurement to zero.

Drug administration

Carbenoxolone (CBX; nacalai tesque, 32775-51) /H₂O, 18beta-Glycyrrhetic acid (18β-GA; abcam, ab142579)/DMSO and (-)-Blebbistatin (FUJIFILM Wako, 021- 17041) were prepared. Time-lapse images were acquired for a single organoid before and after the drug administration. Following the drug adding, organoids were allowed to rest for 30 min to avoid possible effect of turbulence of the medium, and time-lapse images were taken for 5 min.

Three-well hydrogel fabrication

The photoinitiator P2CK was synthesized as previously reported (Li et al., 2013). The target product was verified using proton NMR and Bruker's TopSpin software. Gelatin-Norbornene was synthesized based on previous report (Van Hoorick et al., 2018). The final reaction mixture was transferred to a dialysis tube (5-6 kDa; Spectra Por, cat. no. 132680T) for dialysis against pure water at 40°C for 3 days. After dialysis, the solution's pH was carefully adjusted to 8.0.

A piece of 3-well hydrogel was fabricated on a glass bottom dish (Matsunami) from 100 μL of a solution containing 20 wt % Gelatin-Norbornene, 2 mM of the photoinitiator P2CK, and 20 mM of crosslinker (DTT) (TCI, #D1071) using a two-photon microscope with controllable laser power in 3D space according to the voxel file input (Olympus, in house customized). The voxel files specifying the 3D shape of the hydrogel were designed with Fusion 360 software. (Autodesk).

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Movie 1

Long-term time-lapse imaging after seeding. Time lapse was taken every 2 hours for a total of 100 hours. This video corresponds to **Fig. 1B** [↗](#).

Movie 2

Periodic contractions of the cluster/spheroid on day 3, 5 and 7. Time-lapse images were taken with 700 msec intervals for 5 min. This video corresponds to **Fig. 2A** [↗](#).

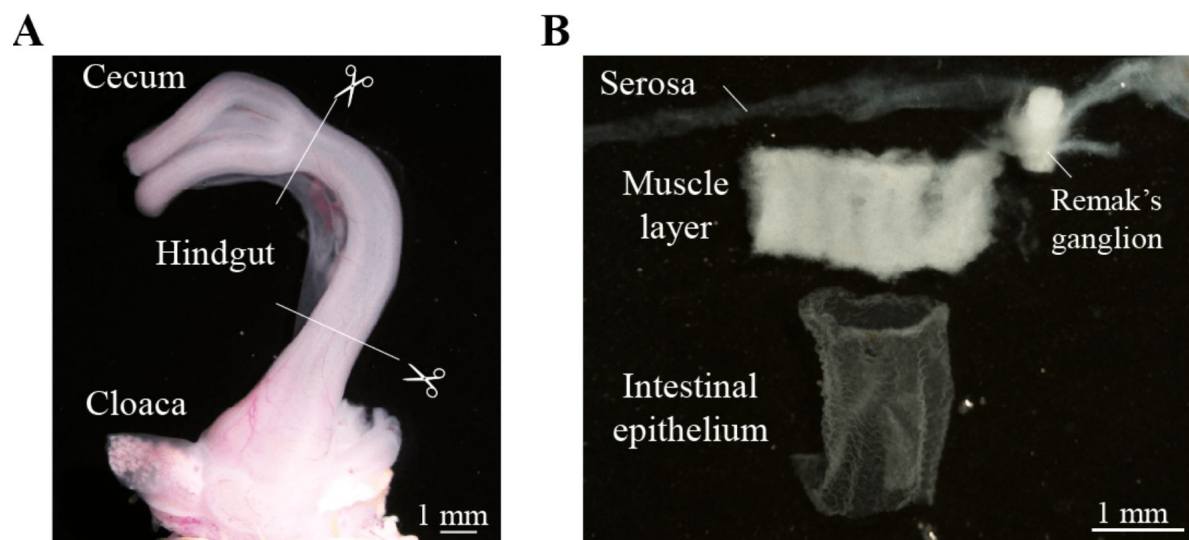
Movie 3

Periodic contractions of a day 7 organoid with CBX. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to **Fig. S3** [↗](#).

Movie 4

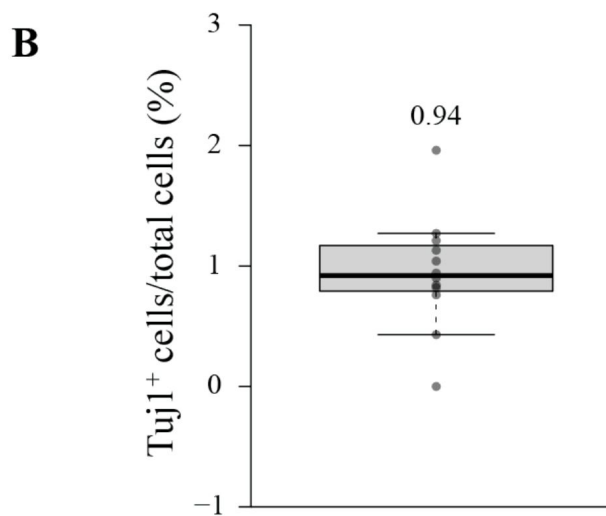
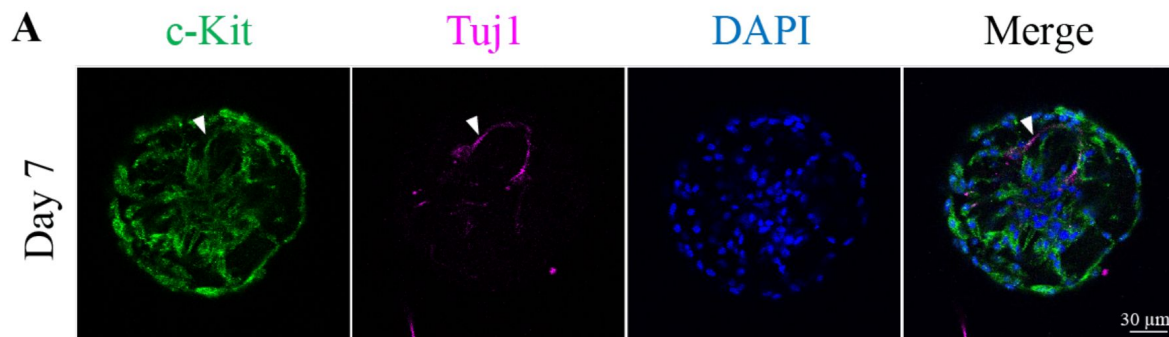
Ca²⁺ dynamics in day 7 gut contractile organoid are concomitant with its contractions. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to **Fig. 4** [↗](#).

Movie 5



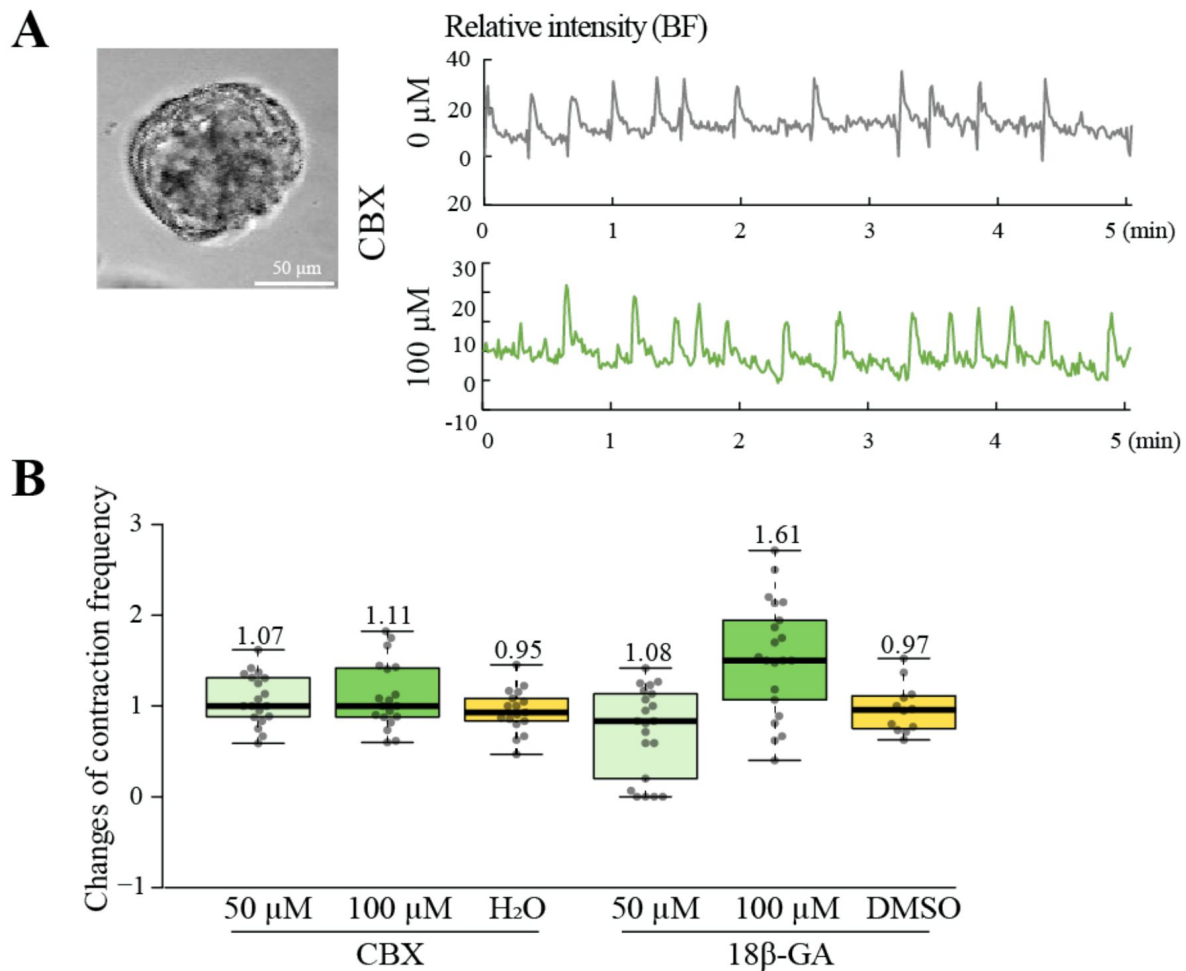
Supplementary Figure S1

(A) Chicken embryonic hindgut at E15. The hindgut was dissected from the bottom of the cecum to the front of the cloaca (white lines). (B) Three layers of E15 hindgut: serosa, muscle layer, intestinal epithelium. Remak's ganglion was also removed. Scale bars: 1 mm in A, B



Supplementary Figure S2

(A) Co-staining of Day 7 organoids with anti-c-Kit- and Tuj1 antibodies. White arrowhead shows a Tuj1⁺ cell. (B) Representation of Tuj1⁺ cells in an organoid at day 7. Scale bar: 30 μ m in A.



Supplementary Figure S3

(A) Day 7 organoid exhibited periodic contractions both before (0 μ M; control) and after adding CBX. Graphs show contractions quantified by the difference in intensity of brightfield in sequential images. (B) Relative changes to the control in contraction frequency in day 7 organoid after CBX. The number is an average. Scale bar: 50 μ m in A.

Ca²⁺ dynamics in a gut contractile organoid with CBX. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to **Fig. 5** [↗](#).

Movie 6

Ca²⁺ dynamics in day 7 gut contractile organoid with Blebbistatin. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to **Fig. 6** [↗](#).

Movie 7

Live imaging during fusion of multiple organoids. Time-lapse images were obtained every 10 minutes for 4 hours. This video corresponds to **Fig. 7A** [↗](#).

Movie 8

Ca²⁺ transients in two gut contractile organoids before and after fusion. Time-lapse images were obtained with 450 msec intervals for 5 min. This video corresponds to **Fig. 7B** [↗](#).

Movie 9

Ca²⁺ transients in three gut contractile organoids in 3-well hydrogel with channels. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to **Fig. 8C** [↗](#).

Movie 10

Ca²⁺ transients in three gut contractile organoids in 3-well hydrogel *without* channels. Time-lapse images were obtained with 450 msec intervals for 5 min. This video corresponds to **Fig. 8G** [↗](#).

Movie 11

Similar assay to movie 10 with 18β-GA administration. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to **Fig. 8H** [↗](#).

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Reviewer #1 (Public Review):

Summary:

In this study, the authors developed an organoid system that contains smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs; pacemaker) but few enteric neurons, and generates rhythmic contractions as seen in the developing gut. The stereotypical arrangements of SMCs and ICCs in the organoid allowed the authors to identify these cell types in the organoid

without antibody staining. The authors took advantage of this and used calcium imaging and pharmacology to study how calcium transients develop in this system through the interaction between the two types of cells. The authors first show that calcium transients are synchronized between ICC-ICC, SMC-SMC, and SMC-ICC. They then used gap junction inhibitors to suggest that gap junctions are specifically involved in ICC-to-SMC signaling. Finally, the authors used an inhibitor of myosin II to suggest that feedback from SMC contraction is crucial for the generation of rhythmic activities in ICCs. The authors also show that two organoids become synchronized as they fuse and SMCs mediate this synchronization.

Strengths:

The organoid system offers a useful model in which one can study the specific roles of SMCs and ICCs in live samples.

Weaknesses:

Since only one blocker each for gap junction and myosin II was used, the specificities of the effects were unclear.

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Reviewer #2 (Public Review):

Summary:

In this study, Yagasaki et al. describe an organoid system to study the interactions between smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs). While these interactions are essential for the control of rhythmic intestinal contractility (i.e., peristalsis), they are poorly understood, largely due to the complexity of and access to the in vivo environment and the inability to co-culture these cell types in vitro for long term under physiological conditions. The "gut contractile organoids" organoids described herein are reconstituted from stromal cells of the fetal chicken hindgut that rapidly reorganize into multilayered spheroids containing an outer layer of smooth muscle cells and an inner core of interstitial cells. The authors demonstrate that they contract cyclically and additionally use calcium imaging to show that these contractions occur concomitantly with calcium transients that initiate in the interstitial cell core and are synchronized within the organoid and between ICCs and SMCs. Furthermore, they use several pharmacological inhibitors to show that these contractions are dependent upon non-muscle myosin activity and, surprisingly, independent of gap junction activity. Finally, they develop a 3D hydrogel for the culturing of multiple organoids and found that they synchronize their contractile activities through interconnecting smooth muscle cells, suggesting that this model can be used to study the emergence of pacemaking activities. Overall, this study provides a relatively easy-to-establish organoid system that will be of use in studies examining the emergence of rhythmic peristaltic smooth muscle contractions and how these are regulated by interstitial cell interactions. However, further validation and quantification will be necessary to conclusively determine show the cellular composition of the organoids and how reproducible their behaviors are.

Strengths:

This work establishes a new self-organizing organoid system that can easily be generated from the muscle layers of the chick fetal hindgut to study the emergence of spontaneous smooth muscle cell contractility. A key strength of this approach is that the organoids seem to contain few cell types (though more validation is needed), namely smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs). These organoids are amenable to live imaging of calcium dynamics as well as pharmacological perturbations for functional assays, and since they are

derived from developing tissues, the emergence of the interactions between cell types can be functionally studied. Thus, the gut contractile organoids represent a reductionist system to study the interactions between SMCs and ICCs in comparison to the more complex *in vivo* environment, which has made studying these interactions challenging.

Weaknesses:

The study falls short in the sense that it does not provide a rigorous amount of evidence to validate that the gut organoids are made of bona fide smooth muscle cells and ICCs. For example, only two "marker" proteins are used to support the claims of cell identity of SMCs and ICCs. At the same time, certain aspects of the data are not quantified sufficiently to appreciate the variance of organoid rhythmic contractility. For example, most contractility plots show the trace for a single organoid. This leads to a concern for how reproducible certain aspects of the organoid system (e.g. wavelength between contractions/rhythm) might be, or how these evolve uniquely over time in culture. Furthermore, while this study might be able to capture the emergence of ICC-SMC interactions as they related to muscle contraction and pacemaking, it is unclear how these interactions relate to adult gastrointestinal physiology given that the organoids are derived from fetal cells that might not be fully differentiated or might have distinct functions from the adult. Finally, despite the strength of this system, discoveries made in it will need to be validated *in vivo*.

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Reviewer #3 (Public Review):

Summary:

The paper presents a novel contractile gut organoid system that allows for *in vitro* studying of rudimentary peristaltic motions in embryonic tissues by facilitating GCaMP-live imaging of Ca²⁺ dynamics, while highlighting the importance and sufficiency of ICC and SMC interactions in generating consistent contractions reminiscent of peristalsis. It also argues that ENS at later embryonic stages might not be necessary for coordination of peristalsis.

Strengths:

The manuscript by Yagasaki, Takahashi, and colleagues represents an exciting new addition to the toolkit available for studying fundamental questions in the development and physiology of the hindgut. The authors carefully lay out the protocol for generating contractile gut organoids from chick embryonic hindgut, and perform a series of experiments that illustrate the broader utility of these organoids for studying the gut. This reviewer is highly supportive of the manuscript, with only minor requests to improve confidence in the findings and broader impact of the work. These are detailed below.

Weaknesses:

(1) Given that the literature is conflicting on the role GAP junctions in potentiating communication between intestinal cells of Cajal (ICCs) and smooth muscle cells (SMCs), the experiments involving CBX and 18Beta-GA are well-justified. However, because neither treatment altered contractile frequency or synchronization of Ca⁺⁺ transients, it would be important to demonstrate that the treatments did indeed inhibit GAP junction function as administered. This would strengthen the conclusion that GAP junctions are not required, and eliminate the alternative explanation that the treatments themselves failed to block GAP junction activity.

(2) Given that 5uM blebbistatin increases the frequency of contractions but 10uM completely abolishes contractions, confirming that cell viability is not compromised at the higher concentration would build confidence that the phenotype results from inhibition of myosin activity. One could either assay for cell death, or perform washout experiments to test for recovery of cyclic contractions upon removal of blebbistatin. The latter may provide access to other interesting questions as well. For example, do organoids retain memory of their prior setpoint or arrive at a new firing frequency after washout?

(3) Regulation of contractile activity was attributed to ICCs, with authors reasoning that Tuj1+ enteric neurons were only present in organoids in very small numbers (~1%). However, neuronal function is not strictly dependent on abundance, and some experimental support for the relative importance of ICCs over Tuj1+ cells would strengthen a central assumption of the work that ICCs the predominant cell type regulating organoid contraction. For example, one could envision forming organoids from embryos in which neural crest cells have been ablated via microdissection or targeted electroporation. Another approach would be ablation of Tuj1+ cells from the formed organoids via tetrodotoxin treatment. The ability of organoids to maintain rhythmic contractile activity in the total absence of Tuj1+ cells would add confidence that the ICCs are indeed the driver of contractility in these organoids.

(4) Given the implications of a time lag between Ca⁺⁺ peaks in ICCs and SMCs, it would be important to quantify this, including standard deviations, rather than showing representative plots from a single sample.

(5) To validate the organoid as a faithful recreation of in vivo conditions, it would be helpful for authors to test some of the more exciting findings on explanted hindgut tissue. One could explant hindguts and test whether blebbistatin treatment silences peristaltic contractions as it does in organoids, or following RCAS-GCAMP infection at earlier stages, one could test the effects of GAP junction inhibitors on Ca⁺⁺ transients in explanted hindguts. These would potentially serve as useful validation for the gut contractile organoid, and further emphasize the utility of studying these simplified systems for understanding more complex phenomena in vivo.

(6) Organoid fusion experiments are very interesting. It appears that immediately after fusion, the contraction frequency is markedly reduced. Authors should comment on this, and how it changes over time following fusion. Further, is there a relationship between aggregate size and contractile frequency? There are many interesting points that could be discussed here, even if experimental investigation of these points is left to future work.

(7) Minor: As seen in Movie 6 and Figure 6A, 5uM blebbistatin causes a remarkable increase in the frequency of contractions. Given the regular periodicity of these contractions, it is a surprising and potentially interesting finding, but authors do not comment on it. It would be helpful to note this disparity between 5 and 10 uM treatments, if not to speculate on what it means, even if it is beyond the scope of the present study to understand this further.

(8) Minor: While ENS cells are limited in the organoid, it would be helpful to quantify the number of SMCs for comparison in Supplemental Figure S2. In several images, the number of SMCs appears quite limited as well, and the comparison would lend context and a point of reference for the data presented in Figure S2B.

(9) Minor: additional details in the Figure 8 legend would improve interpretation of these results. For example, what is indicated in orange signal present in panels C, G and H? Is this GCAMP?

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Author response:**Generals:**

We deeply appreciate the efforts by the Senior and Reviewing Editors, and also thank the three reviewers for their careful reading of the MS and their constructive comments, which are very helpful to improve our MS. We agree that we extend our efforts to elaborate the pharmacological analyses including clarification of the penetrance of GAP junction inhibitor(s), and effectiveness and specificity of the drugs. We plan to test at least L-type calcium channel blocker nifedipine. Concerning the reproducibility of the phenotypes, we indeed repeated experiments at multiple times for each of the analyses. While we demonstrated in the current version a series of representative data for simplicity along with explanation in the text that we conducted multiple times of experiments, in a revised version we will improve the demonstration so that readers/reviewers can be convinced with the reproducibility of the data. We will also try to test other markers to look into cell types constituting the gut contractile organoid

Specifics:

Our provisional responses to “The weakness” raised by the reviewers are as follows:

Reviewer #1:

Please see the responses shown above (“Generals”).

Reviewer #2:

In addition to the responses in “Generals”, our response also includes the followings: We will look into wavelength between contractions/rhythm of the organoid. We agree that our organoids derived from embryonic hind gut (E15) might not necessarily recapitulate the cell function in adult. However, it has well been accepted in the field of developmental biology that studies with embryonic tissue/cells make a huge contribution to unveil how complicated physiological cell function is underpinned. Nevertheless, we will carefully consider in the revised version so that the MS would not send misleading messages. Recent advances have also shown that 3D organoids can somehow “replace/substitute for” a complicated in vivo specimen when a particular cellular function is a focus of study.

Reviewer #3:

We appreciate a strong support of our findings.

(1) We plan to perform positive control experiments, for example, to test if the drugs we use would interfere cardiac muscle functions.

(2) We plan to do wash-out experiment to confirm 10uM blebbistatin does not kill the cells. Thank you for this suggestion.

(3) We plan to conduct tetrodotoxin treatment. Since experiments with such toxic reagents are not encouraged by our institute, we will perform experiments with a necessary-minimum amount.

(4) We plan to address this point properly

1. It is well predictable that blebbistatin would stop the gut movement in an explanted hindgut, and it is also well established that gut contractions (movements) are concomitant with Ca^{2+} transients. It would indeed be interesting to see how GJ inhibitors affect such in vivo gut movement. However, since all the reviewers and the Reviewing Editor pointed out, sensitivity (concentration) and penetrance of the drug is an important point of concern, we think that the in vivo analyses will be a next step to go in near future.

(6) We have indeed noticed that contraction frequency is reduced after organoidal fusion. It seems as if cells communicate with each other to decide which rhythm they need to be adjusted to. Furthermore, contraction frequency tends to be slow down when the organoid becomes larger in size. It might be attributed to a delay in conductance between cells over growing distance. We plan to either quantify these potentially interesting phenomena or make a concise speculation in the revised version.

(7)-(10) Thank you for these comments. We will fix them.

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